Syntax and Semantics

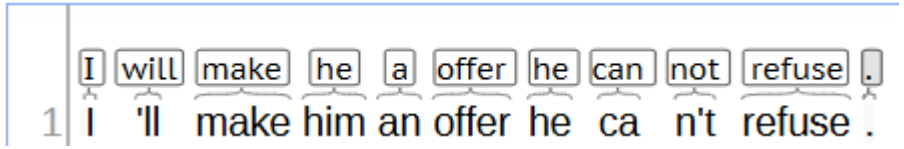
References:

1. Bender: Linguistic Fundamentals for NLP I
2. Bender and Lascarides: Linguistic Fundamentals for NLP II
3. Asher, Kumari: Malayalam
4. Wikipedia
5. Nan Decker. 1985. The use of syntactic clues in discourse processing.

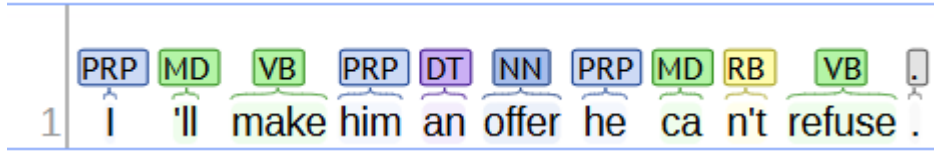
Introduction

- *I'll make him an offer he can't refuse.*

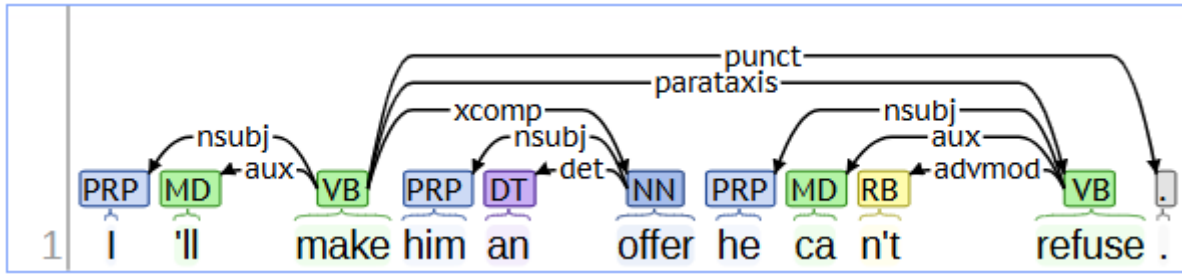
Lemmas



Parts-of-speech

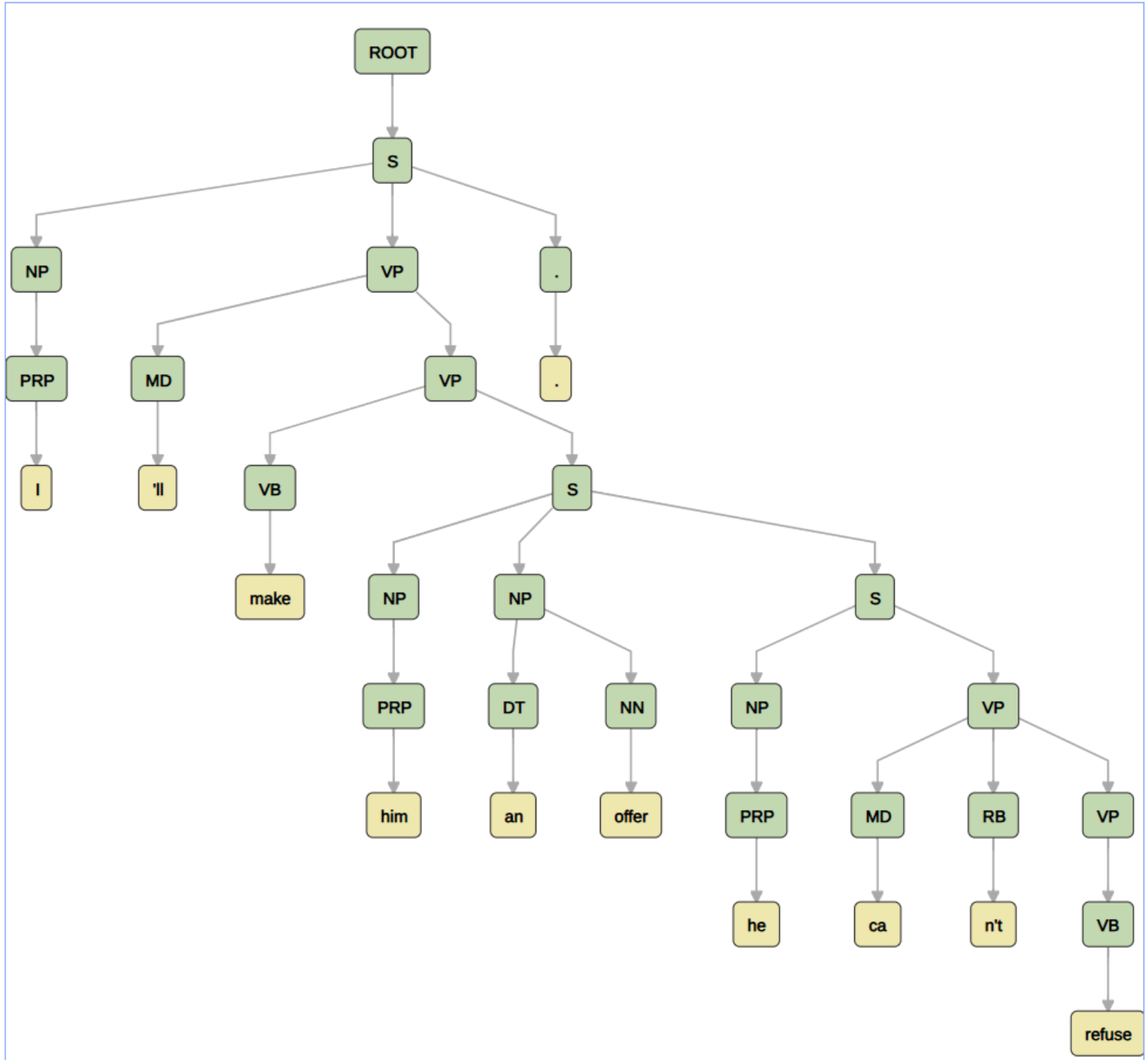


Dependency parse



https://corenlp.run/

Constituency Parse:



https://corenlp.run/

Parts-of-speech

- ADJ: adjective (big, third, incomprehensible)
- ADP: adposition (in, to, during)
- ADV: adverb (very, tomorrow, when, there, nowhere)
- AUX: auxiliary (has, was, should, is)
- CCONJ: coordinating conjunction (and, or, but)
- DET: determiner (a, an, the, this, which, no)
- INTJ: interjection (yep, ohh)
- NOUN: noun (girl, tree, decision)
- NUM: numeral (27, nine, hundred)
- PART: particle (‘s, ‘nt)
- PRON: pronoun (he, she, your, who, somebody)
- PROPN: proper noun (Mary, Mumbai, Matrubhumi)
- PUNCT: punctuation
- SCONJ: subordinating conjunction (that, if, while)
- SYM: symbol
- VERB: verb (run, ran, running)
- X: other

https://universaldependencies.org/u/pos/index.html

Constituents

- Words within sentences form intermediate groupings called constituents
- Kim [read a book], [gave it to Sandy], and [left].
- Kim read [[a very interesting book] [about grammar]].
- Kim read [it].
- A syntactic head determines the internal and external dependencies
- Kim planned to give Sandy books.
- [**Kim**] [**planned** to give] [**Sandy**] [her physics **books**, which] [she had **purchased**] [from a **bookstore**] [in Chandani **Chowk**].

Dependencies

Table 8.1: Argument dependency types from de Marneffe and Manning 2011

Tag	Type	Example
nsubj	nominal subject	*Kim* **took** the picture.
nsubjpass	passive nsubj	*The picture* was **taken** by Kim.
csubj	clausal subject	*What she said* **makes** sense.
csubjpassive	passive csubj	*That she lied* was **suspected** by everyone.
xsubj	controlling subject	*Kim* likes to **take** pictures.
agent	in passives	The picture was **taken** by *Kim*.
expl	existential *there*	*There* **is** a ghost in the room.
dobj	direct object	They **win** *the lottery*.
iobj	indirect object	She **gave** *me* a raise.
ccomp	clausal complement	He **says** *that you like to swim*.
xcomp	controlled clause	You **like** *to swim*.
acomp	adjectival complement	Bill **looks** *very big*.
pcomp	preposition's clausal complement	They heard **about** *you missing class*.
pobj	object of preposition	They sat **on** *the chair*.
prt	phrasal verb particle	They **shut** *down* the station.
cop	complement of copular verb	Bill **is** *big*.
attr	*wh* complement of copular verb	*What* **is** that?

Dependencies to Arguments

a. Kim gave Sandy [the book….].
b. Kim gave [the book….] to Sandy.

c. Kim sent Sandy the letter.
d. Kim sent the letter to Sandy.

e. Kim threw Sandy [a party at….].
f. Kim threw [a party at….] for Sandy.

g. Kim baked Sandy [a cake with ….].
h. Kim baked [a cake with ….] for Sandy.

Arguments and Functions

- Thematic roles
  - Agent
    - A participant who does or causes something (represented by a verb) possibly intentionally
    - Subjects of: *eat, kill,* ~~*love*~~
  - Patient
    - A participant who is affected by something
    - Objects of: *eat, kill,* ~~*love*~~
  - Experiencer
    - A participant who experiences or is aware of something
    - Subjects of: *love;* Object of: *annoy*
  - Theme
    - A participant who undergoes a change because of something
    - Objects of: *give, make;* Subjects of: *die, walk*

VerbNet

Go To Comments

destroy-44

Members: 31, Frames: 3

Members

ANNIHILATE (FN 1; WN 1)	DEVASTATE (FN 1; WN 1; G 1)	MUTILATE
BLIGHT (WN 1)	DISFIGURE (WN 1)	OBLITERAT
BLITZ (WN 1)	EFFACE (WN 3; G 3)	RACK (WN
DAMAGE (FN 1; WN 1; G 1)	EXTERMINATE (FN 1; WN 1; G 1)	RAVAGE (W
DECIMATE (WN 2)	EXTIRPATE (WN 1)	RAZE (FN 1
DEMOLISH (FN 1; WN 1; G 1)	LEVEL (FN 1)	RUIN (FN 1
DESECRATE (WN 1, 2)	LOUSE_UP	SHATTER (
DESTROY (FN 1; WN 1, 2; G 1)	MAIM (WN 1)	SHIPWREC

Roles

- Agent [+int_control]
- Patient [+concrete]
- Instrument [+concrete]

Frames

NP V NP

EXAMPLE	"The Romans destroyed the city."
SYNTAX	Agent V Patient
SEMANTICS	cause(Agent, E) destroyed(result(E), Patient)

NP V NP PP.instrument

EXAMPLE	"The builders destroyed the warehouse with explosives."
SYNTAX	Agent V Patient {with} Instrument
SEMANTICS	cause(Agent, E) use(during(E), Agent, Instrument) destroyed(result(E), Patient)

https://verbs.colorado.edu/verb-inde

Selectional Preferences

- He can read.
- He read the newspaper.
- He read Marquez in college.
- He read her mind.
- He read the delicious food.
- Kumar bought a pen and a book. **It** was a pirated copy of Thinking Fast and Slow.

Metaphors

- *A **metaphor** is a figure of speech that describes an object or action in a way that isn't literally true, but helps explain an idea or make a comparison.… Metaphors are used in poetry, literature, and anytime someone wants to add some color to their language.*

Melanie Mitchell, 2019

Metaphors We Live By

- Time is money
  - *You are **wasting** my time*
  - *I **spent** a lot of time doing the homework*
- Up/Down and High/Low
  - *That **boosted** my confidence*
  - *India is 3 **down***
  - *I am feeling bit **low** today*
  - *Her success graph is going **up***
- Mind/Relation as a machine
  - *He was down after **break** up with that girl*
  - *The experience **shattered** him*
- Bitter/Sweet as Bad/Good
  - I had bitter memories of working there
  - She is so sweet!

Lakoff and Johnson, 1980

Black and White

- Black is bad and White is good

  —— with gray in between

- Dark is bad and bright is good
- Morning, evening and night

Characteristics of informativeness

- Clause
- Assertion: expression of belief
- Tense and aspect
- Transitivity
  - an agent is acting on a patient that is directly affected by this action (energy flows from agent to patient)
    - `Kumar killed Kumari`
    - `Kumar loves Kumari`
    - `Kumar drank the milk`
    - `Kumar drank some milk`
- Voice
- Nominalization

https://www.oxfordbibliographies.com/display/document/obo-9780199772810/obo-9780199772810-0148.xml

Clause

- A clause is a group of words that contains a subject and a verb that have a relationship
- A sentence can have one or more clauses
  - **She quickly fixed the sink.**
  - **She fixed the sink** without facing difficulty.
  - **My friend** who rides a motorcycle **said it's a great weekend to ride.**
- Types
  - Independent or main clause
  - Dependent or subordinate clause
  - Relative clause (modifies a noun or noun phrase)
  - Restrictive and non-restrictive
    - The astronaut **who first stepped on the moon** was Neil Armstrong.
    - The cat was allowed on the bed, **which annoyed the dog.**

https://www.grammarly.com/blog/commonly-confused-words/using-that-and-which-is-all-about-restrictive-and-non-restrictive-clauses/

Informativeness of clause

- Non-restrictive clause: *background*
- Appositives: *background*
  - `Hermione Granger, a witch at Hogwarts School, is accomplished at spells.`
- Main clause: *foreground*

https://www.grammarly.com/blog/punctuation-capitalization/appositive/

Morphosyntax of Verbs

- Tense
  - grammaticalized location in time (past, present, future)
  - *She purchased a book*
- Aspect
  - grammaticalization of expression of internal temporal consistency (unitary, continuous)
  - *She had been purchasing books every month*

Examples

	Past	Present	Future
Simple	*I **helped** my neighbor yesterday.*	*I **help** my neighbor every day.*	*I **will help** my neighbor tomorrow.*
Perfect	*I **had helped** my neighbor clean his attic before I fixed his car.*	*I **have helped** my neighbor too much this week.*	*I **will have helped** my neighbor a hundred times by the end of the month.*
Continuous	*I **was helping** my neighbor when he brought me iced tea.*	*I **am helping** my neighbor while he fixes up his house.*	*I **will be helping** my neighbor next month when he moves.*
Perfect continuous	*I **had been helping** my neighbor for a year before he finally thanked me.*	*I **have been helping** my neighbor since I moved in.*	*I **will have been helping** my neighbor for a year next month.*

https://www.grammarly.com/blog/parts-of-speech/verb-tenses/

Types of times

- Speech Time, Event Time and Reference Time

(a) I have lost my watch.

(b) Yesterday, I lost my watch.

(c) but I have found it again

(d) Chapman breathed a sigh of relief.

(e) When Nixon was elected, Chapman breathed a sigh of relief.

https://plato.stanford.edu/entries/tense-aspect/

Reichenbach, H., 1947, Elements of Symbolic Logic, New York: Dover Publications, Inc.

Transitivity

- Kinetic
  - `throw`
  - `to be`
  - `walk?`
  - `stood?`
- Telic: action with a natural end point
  - `Kumar was making chair`
  - `Kumar was reading`
- Punctual: difficult to decompose into sub-actions
  - `Kumar kicked the ball`
  - `Kumar studied for 4 hours`
- Volitional: intentionality of an action
  - `Kumar slipped on the wet floor`
- Affirmative
  - `Kumar ate one pizza`
  - `Kumar did not eat a slice of pizza`
- Realis: actuality of an action
  - `Kumar should read those books`
  - `Kumar read those books`

Agents

- Proper nouns
- Humans
- Animate entities
- Concrete
- Singular
- Count

Nominalization

- Converting verb to noun
  - *elect ---> election*
  - *nominate ---> nomination*
- Election: a formal group decision-making process whereby a population chooses an individual or multiple individuals to hold public office.

U.S. Vice President Kamala Harris, endorsed by Joe Biden to be a new Democratic presidential nominee.

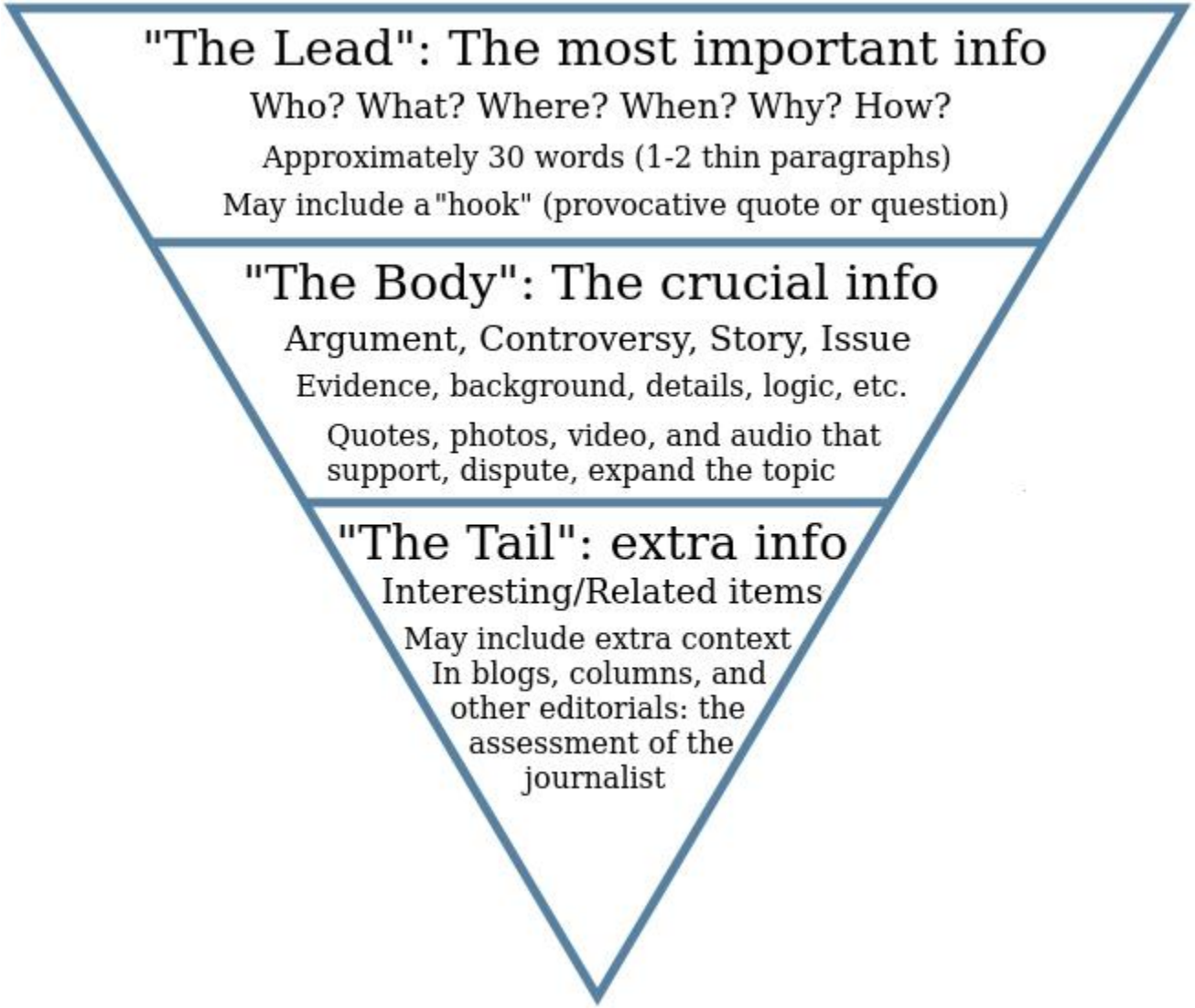
She said that she intends to "earn and win" the **nomination** of the party and unite the nation to defeat her Republican rival Donald Trump in upcoming **election** in November 2024.

https://en.wikipedia.org/wiki/Election

News Articles: Syntax and Discourse

- Information is presented in descending order of importance
- Paragraphs as blocks: no link to adjacent paragraphs
- More than one episodes:
    1. Chronology of the main events
    2. References to the related past and possible future events
    3. Texture: division of material as *foreground vs background*
    4. Text, context and texture

    Texture in written text arises as the writer attempts to monitor and control the flow of information through the text in a manageable way so that the reader is guided towards the kind of interpretation intended by the writer.



https://ohiostate.pressbooks.pub/stratcommwriting/chapter/inverted-pyramid-style/
https://journals.equinoxpub.com/index.php/books/article/viewFile/BOOK-209-520-4/33980

Characteristics of informativeness

- Clause
  - main clause
- Assertion: expression of belief
  - positive assertion
- Tense and aspect
  - simple past
- Transitivity
  - high transitivity verbs with strong agents
  - nominalization has low transitivity
- Voice
  - active

https://www.oxfordbibliographies.com/display/document/obo-9780199772810/obo-9780199772810-0148.xml

Applications

- Summarization
- Ordering of events

Extensions

- Wikipedia articles
- Scientific documents